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# GPS/INS Kalman Filter Design for Spacecraft Operating in the Proximity of the International Space Station

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**The next generation reusable launch vehicle will operate in five flight phases: ascent, on-orbit, proximity operations, re-entry and landing. Navigation during each of these flight phases presents unique challenges. The Space Shuttle addresses these challenges by the use of a number of navigation sensors. However, an integrated GPS/INS navigation system may be able to meet the navigation requirements of all flight phases. Integrated GPS/INS systems have been built and demonstrated for the ascent, re-entry and landing phases and their performance is known. However, the same cannot be said for integrated GPS/INS systems for the on-orbit and proximity operations flight phases. Therefore, this research examines the performance of GPS/INS navigation during a rendezvous with the ISS. Error models for INS and GPS navigation sensors operating in the vicinity of the ISS have been developed. The GPS error model includes the effects of GPS signal blockage and multipath near the ISS. These error models have been used to develop an integrated GPS/INS extended Kalman filter. A simulation of the filter has been developed and the first test case shows position errors of less than 1 meter and velocity errors of less than 0.02 m/s in each axis. The second test case shows the position errors grow to approximately 47.5 meters and the velocity errors grow to 0.13 m/s during a ten minute GPS outage. Finally, the third test case shows that the GPS/INS EKF performance is not significantly degraded by GPS signal blockage due to the ISS during a simulated rendezvous with the ISS.**

## Introduction

The goal of NASA's Space Launch Initiative (SLI) program is to advance the technologies needed for the next generation reusable launch vehicle (RLV). The next generation RLV will operate in five flight phases: ascent, on-orbit, proximity operations, re-entry and landing. Navigation during each of these flight phases presents unique challenges. The Space Shuttle addresses these challenges by the use of a number of navigation sensors, many of which are used only during one flight phase. However, it has been hypothesized that an Inertial Navigation System (INS) integrated with a Global Positioning System (GPS) receiver may be able to meet the navigation requirements of all flight phases.

A GPS-only navigation system is not adequate because GPS is subject to outages due to blockage, atmospheric ionization during re-entry and delta-v maneuvers. GPS is also subject to errors and integrity problems due to multipath and other sources. For example, when a spacecraft approaches the International Space Station (ISS) to perform rendezvous and docking, the signals from the GPS satellites may be blocked by the ISS or degraded by multipath.

Integrated GPS/INS systems have been analyzed for the re-entry and landing phases by Braden, Browning and Gelderloos.[1] GPS/INS systems have also been demonstrated to be capable of precision landing. However, the same cannot be said for the on-orbit and proximity ops flight phases. Automated rendezvous and docking has been identified by the SLI program as an area requiring further research and development. Currently, the Space Shuttle uses a partially manual system for rendezvous, but a fully automated system could be safer and more reliable.[2] Therefore, this research examines GPS/INS navigation during a rendezvous with the ISS.

### Previous Work

The Space Integrated GPS/INS (SIGI) is an example of a spaceborne integrated GPS/INS system.[3] SIGI is used on the ISS and was planned to be used on the Crew Return Vehicle (CRV). However, it is not known whether SIGI will meet the navigation requirements for all flight phases or whether it will meet the size, weight, and power requirements for the next generation RLV. In addition, the GPS and INS in SIGI are integrated in a loosely coupled architecture so it may be possible to improve its navigation performance by changing to a more tightly integrated design.

Much work has been done on integrated GPS/INS systems for aircraft and missiles. There are many examples of these systems in operation today, such as the Honeywell H764G Embedded GPS/INS, which is currently being flown in several military aircraft and

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has been shown to be capable of precision approach and landing by Elchynski.[4]

Ebinuma demonstrated the use of GPS for rendezvous. He developed an extended Kalman filter to perform real-time relative navigation and used a hardware-in-the-loop test facility, which integrates navigation and guidance for rendezvous. However, his models assumed that GPS was always available and did not include multipath or blockage of the GPS signals.[5]

The most recent research on navigation during proximity operations conducted by Um examined spacecraft relative navigation using an integrated GPS/INS in the vicinity of the ISS.[6] The GPS/INS was a loosely coupled system, which included two Kalman filters, one to process the GPS measurements and the other to combine the INS measurements with the output of the GPS filter. The INS part of the system was simulated in software and the GPS measurements were collected from a Mitel Architect GPS receiver being stimulated by a GSSI STR4760 GPS simulator.

The major contributions of this paper include:

1. A spacecraft INS error model is developed. Most INS error models are designed for terrestrial navigation and therefore are formulated in an earth-fixed reference frame. The error model developed in this paper is formulated in an inertial reference frame, which is more useful for spacecraft in orbit.

2. A GPS error model which includes the effects of GPS signal blockage and multipath caused by the ISS.

3. A proposed GPS/INS Kalman filter design for an orbiting spacecraft and a simulation of the navigation performance during a rendezvous with the ISS.

## INS Error Model

In order to design an integrated GPS/INS navigation system for spacecraft, a linear INS error model has been developed. The INS system-level error model describes the errors in position, velocity and attitude due to various error sources. In most cases, these models are formulated in terms of latitude, longitude and altitude errors and east, north and vertical velocity errors. Attitude errors have been expressed in terms of rotation matrices, tilt angles or quaternions.

In the case of a spacecraft in orbit, it is convenient to formulate the INS system-level error model in an inertial reference frame as Friedland did in 1978.[7] Friedland's formulation is used in this paper. A similar model was developed and used by Crain and Bishop for analyzing Mars entry navigation.[8]

The sensor-level models describe the errors in specific force measurements taken by accelerometers and errors in angular rate measurements taken by gyroscopes. The sensor error models used by most authors have the same form but sometimes differ in the stochastic processes (random walk or Gauss-Markov processes) used to model various component errors such as accelerometer bias or gyro drift. These differences are due to the nature of the instrument or the data available on the instrument. Detailed data on a particular sensor are often not available so a simpler model must be used.

### Concept of Inertial Navigation

The fundamental equation of inertial navigation comes directly from Newton's Second Law:

$$\frac{d^2\underline{\mathbf{r}}}{dt^2} = C_B^I \underline{\mathbf{f}} + \underline{\mathbf{g}} \tag{1}$$

where $\underline{\mathbf{r}}$ is the inertial position vector, $C_B^I$ is the matrix which transforms a vector from the INS body frame (B) to the inertial reference frame (I), $\underline{\mathbf{f}}$ is the total specific force vector, and $\underline{\mathbf{g}}$ is the acceleration due to gravity. This equation is integrated once to obtain velocity and a second time to obtain position, both relative to the inertial reference frame. Accelerometers measure specific force, which does not include forces due to gravity. Gyroscopes (gyros) measure the angular motion of the INS body with respect to inertial space in order to compute $C_B^I$. An INS uses an internal gravity model to estimate the acceleration due to gravity based on its current estimate of position and attitude.

The major sources of error in an INS include initialization errors, system alignment errors, sensor errors, gravity model errors and quantization and computational errors. These errors are characterized in the INS error model.

The gyros and accelerometers are fixed to the body of the vehicle in a strapdown INS. Therefore, the outputs of the accelerometers and gyros are assumed to be provided in the body frame with small misalignments as described in the INS error model.

### INS System-Level Error Model

The navigation errors committed by an inertial navigation system can be modelled by a set of first order, linear ordinary differential equations. Define the error state vector $\delta\underline{\mathbf{x}}$ as:

$$\delta\underline{\mathbf{x}} = \begin{bmatrix} \delta\underline{\mathbf{r}} \\ \delta\underline{\mathbf{v}} \\ \delta q \end{bmatrix} \tag{2}$$

where $\delta\underline{\mathbf{r}}$ is the position error, $\delta\underline{\mathbf{v}}$ is the velocity error and $\delta q$ represents the error in the platform to inertial reference frame quaternion. Then the INS system-level error model is:[7]

$$\begin{bmatrix} \delta\dot{\underline{\mathbf{r}}} \\ \delta\dot{\underline{\mathbf{v}}} \\ \delta\dot{q} \end{bmatrix} = \begin{bmatrix} \delta\underline{\mathbf{v}} \\ C_B^I(\hat{q})\left[2\bar{A}^T\bar{R}^T\delta q + \delta\underline{\mathbf{f}}\right] + \frac{\partial\underline{\mathbf{g}}(\underline{\mathbf{r}})}{\partial\underline{\mathbf{r}}}\delta\underline{\mathbf{r}} \\ Q(\hat{q})\,\delta\underline{\omega} + \Omega(\hat{\underline{\omega}})\,\delta q \end{bmatrix}$$

where, $\omega_1$, $\omega_2$ and $\omega_3$ are the body frame compo-

nents of the instantaneous angular velocity vector and:

$$\Omega\left(\underline{\omega}\right) = \frac{1}{2}\begin{bmatrix} 0 & \omega_3 & -\omega_2 & \omega_1 \\ -\omega_3 & 0 & \omega_1 & \omega_2 \\ \omega_2 & -\omega_1 & 0 & \omega_3 \\ -\omega_1 & -\omega_2 & -\omega_3 & 0 \end{bmatrix} \tag{3}$$

and

$$Q\left(q\right) = \frac{1}{2}\begin{bmatrix} q_4 & -q_3 & q_2 \\ q_3 & q_4 & -q_1 \\ -q_2 & q_1 & q_4 \\ -q_1 & -q_2 & -q_3 \end{bmatrix} \tag{4}$$

and

$$\bar{R} = \begin{bmatrix} 2Q & -q \end{bmatrix} \tag{5}$$

and

$$\bar{A} = \begin{bmatrix} 0 & -f_3 & f_2 \\ f_3 & 0 & -f_1 \\ -f_2 & f_1 & 0 \\ -f_1 & -f_2 & -f_3 \end{bmatrix} \tag{6}$$

The gravity gradient matrix, $\underline{G} = \frac{\partial \mathbf{g}(\underline{\mathbf{r}})}{\partial \underline{\mathbf{r}}}$ for a two-body gravity field is:

$$\underline{G}_{3\times3} = \begin{bmatrix} \frac{3\mu x^2}{r^5} - \frac{\mu}{r^3} & \frac{3\mu xy}{r^5} & \frac{3\mu xz}{r^5} \\ \frac{3\mu xy}{r^5} & \frac{3\mu y^2}{r^5} - \frac{\mu}{r^3} & \frac{3\mu yz}{r^5} \\ \frac{3\mu xz}{r^5} & \frac{3\mu yz}{r^5} & \frac{3\mu z^2}{r^5} - \frac{\mu}{r^3} \end{bmatrix} \tag{7}$$

The INS system-level error equations can also be written as follows:

$$\begin{bmatrix} \delta\dot{\underline{\mathbf{r}}} \\ \delta\dot{\underline{\mathbf{v}}} \\ \delta\dot{q} \end{bmatrix} = \begin{bmatrix} 0_{3\times3} & I_{3\times3} & 0_{3\times3} \\ \underline{G}_{3\times3} & 0_{3\times3} & D_{3\times4} \\ 0_{4\times3} & 0_{4\times3} & \Omega\left(\hat{\underline{\omega}}\right) \end{bmatrix} \begin{bmatrix} \delta\underline{\mathbf{r}} \\ \delta\underline{\mathbf{v}} \\ \delta q \end{bmatrix} + \begin{bmatrix} 0_{3\times3} & 0_{3\times3} \\ 0_{3\times3} & C_B^I\left(\hat{q}\right) \\ Q\left(\hat{q}\right) & 0_{4\times3} \end{bmatrix} \begin{bmatrix} \delta\underline{\omega} \\ \delta\underline{\mathbf{f}} \end{bmatrix} \tag{8}$$

where $D = 2C_B^I\left(\hat{q}\right)\bar{A}^T\bar{R}^T$. The vectors $\delta\underline{\omega}$ and $\delta\underline{\mathbf{f}}$ represent the errors in the gyro angular velocity and accelerometer specific force measurements. These vectors are obtained from the gyro and accelerometer error models described below.

**Gyro Error Model**

The purpose of the gyro error model is to provide the $\delta\underline{\omega}$ term in equation (8). The specific details of any gyro error model depend on the type of gyro and the characteristics of the errors found during testing. Gyro manufacturers usually provide some specification numbers but do not publish the detailed error models. Therefore, early in the navigation system design process, before a particular gyro has been selected, simplified error models are used to perform trade studies and requirements analysis.

The error model for a set of three orthogonal gyros can be defined by:[9]

$$\delta\underline{\omega} = \begin{bmatrix} \delta SF_{Gx} & -XG_z & XG_y \\ YG_z & \delta SF_{Gy} & -YG_x \\ -ZG_y & ZG_x & \delta SF_{Gz} \end{bmatrix} \underline{\omega} + \delta b_G + \delta nl_G + n_G$$

where the gyro scale factor error vector is defined as:

$$\delta SF_G = \begin{bmatrix} \delta SF_{Gx} & \delta SF_{Gy} & \delta SF_{Gz} \end{bmatrix}^T$$

the vector of gyro misalignments is defined as:

$$\delta MA_G = \begin{bmatrix} XG_z & XG_y & YG_z & YG_x & ZG_y & ZG_x \end{bmatrix}^T$$

$\delta b_G$ represents the gyro bias, $\delta nl_G$ represents the gyro g-sensitive errors and $n_G$ represents the gyro measurement noise. Gyro scale factor and alignment errors are modelled as Gaussian random constants.

There are two components in gyro drift. There is the gyro drift (also referred to as bias repeatability or bias stability) that varies from turn-on to turn-on. This is modelled as a Gaussian random constant. The second part is bias drift or variability after turn-on (sometimes referred to as in run stability) and is usually modelled as a random walk or first order Markov process.

Gyro measurement noise is usually the predominant error source in ring laser and fiber optic gyros but is usually not included in error models for most mechanical gyros. This is modelled as zero-mean, white Gaussian noise and causes angular random walk.

**Accelerometer Error Model**

The purpose of the accelerometer error model is to provide the $\delta\underline{\mathbf{f}}$ term in equation (8). The accelerometer error model is very similar to the gyro error model. The error model for a set of three orthogonal accelerometers can be defined by:[9]

$$\delta\underline{\mathbf{f}} = \begin{bmatrix} \delta SF_{Ax} & -XA_z & XA_y \\ YA_z & \delta SF_{Ay} & -YA_x \\ -ZA_y & ZA_x & \delta SF_{Az} \end{bmatrix} \underline{\mathbf{f}} + \delta b_A + \delta nl_A + n_A$$

where the accelerometer scale factor error vector is defined as:

$$\delta SF_A = \begin{bmatrix} \delta SF_{Ax} & \delta SF_{Ay} & \delta SF_{Az} \end{bmatrix}^T$$

the vector of accelerometer misalignments is defined as:

$$\delta MA_A = \begin{bmatrix} XA_z & XA_y & YA_z & YA_x & ZA_y & ZA_x \end{bmatrix}^T$$

$\delta b_A$ represents the accelerometer bias, $\delta nl_A$ represents the accelerometer non-linearity and $n_A$ represents the accelerometer measurement noise. Accelerometer scale factor and misalignment errors are modelled as Gaussian random constants.

Accelerometer bias error is modelled as a random constant plus a random walk or first order Markov process. Accelerometers may exhibit errors that are linear and quadratic functions of acceleration. These errors are modelled as Gaussian random constants. Accelerometer measurement noise is modelled as white Gaussian noise and causes velocity random walk.

## GPS Measurement Model

A model of the measurements obtained by a GPS receiver operating in the vicinity of the ISS is needed to quantify GPS navigation performance. The GPS measurement model includes a model of the GPS constellation, a range measurement error model, a GPS signal blockage model and a multipath model.

### GPS Constellation Model

A model of the GPS constellation was constructed from a daily global broadcast ephemeris file in RINEX format from the National Geodetic Survey CORS data website. This file contains the GPS broadcast ephemeris parameters for each satellite in the constellation for March 1, 2001. There were a total of 29 satellites in the constellation. The ephemeris parameters and the equations used to determine the positions of the GPS satellites at a given time are described in GPS-ICD-200.[10] This GPS constellation model was used for all simulations in this study.

### Range Measurement Error Model

A simplified GPS range measurement error model is used in this study. GPS satellite clock bias, ephemeris error, ionospheric and tropospheric effects are assumed to be negligible. Therefore, the range error for the $j$th satellite at time $t_i$ can be expressed as:

$$\delta\rho_{j-GPS}(t_i) = (c - \dot{\rho})\delta t_R(t_i) + MP_j(t_i) + v_j(t_i) \quad (9)$$

where $c$ is the speed of light, $\dot{\rho}$ is the range-rate, $\delta t_R(t_i)$ is the user clock bias, $MP_j(t_i)$ is the range error due to multipath, and $v_j(t_i)$ is white Gaussian noise representing receiver measurement noise. GPS satellite clock bias and ephemeris errors have not been included but will be added to the model in the future. GPS C/A code measurements are used in this study.

### Receiver Clock Bias Model

The receiver clock model presented in Ebinuma[5] is used in this study. The receiver clock bias and drift are modelled as random walk processes. The receiver clock model is:

$$\begin{aligned} \dot{b_c}(t) &= d_c(t) + w_f(t) + hc \\ \dot{d_c}(t) &= w_g(t) \end{aligned} \quad (10)$$

where $b_c$ is the clock bias scaled to meters ($b_c = c\delta t_R$), $d_c$ is the clock drift, $h$ is the frequency correction due to relativistic effects, and $w_f$ and $w_g$ are zero mean, white Gaussian noise with the power spectral amplitudes of $S_f$ and $S_g$, respectively.

In near circular orbits, the relativistic effect can be approximated by:

$$h = \frac{1}{c^2}\left(\frac{\mu}{R_E} - \frac{\mu}{r} - \frac{v^2}{2}\right) \quad (11)$$

where $R_E$ is the radius of the Earth, $r$ is the distance of the receiver from the center of the Earth and $v$ is the inertial velocity of the receiver.

The discrete time process noise covariance associated with the receiver clock drift and bias is given by:

$$Q = \begin{bmatrix} S_f\Delta t + S_g\frac{\Delta t^3}{3} & S_g\frac{\Delta t^2}{2} \\ S_g\frac{\Delta t^2}{2} & S_g\Delta t \end{bmatrix} \quad (12)$$

where $\Delta t$ is the time between samples.

Typical power spectral amplitudes of temperature controlled crystal oscillators commonly used on commercial GPS receivers are:

$$\begin{aligned} S_f &= 1.0c^2 \times 10^{-19}(m^2/s) \\ S_g &= 4.0c^2\pi^2 \times 10^{-20}(m^2/s^3) \end{aligned}$$

### ISS Signal Blockage Model

It has been hypothesized that the ISS will block GPS signals needed by other spacecraft to navigate during rendezvous operations. To analyze this effect, the GPS signal blockage due to the ISS is modelled as sphere centered at the ISS position with diameter, $d = 100$ meters.

The line of sight vector from the satellite to the $j$th GPS satellite is found by:

$$\underline{\mathbf{r}}_{LOS_j} = \underline{\mathbf{r}}_{GPS_j} - \underline{\mathbf{r}} \quad (13)$$

The GPS antenna is assumed to be pointed along the $\underline{\mathbf{r}}$ vector, therefore, the declination angle ($\theta$) between the antenna boresight and the line of sight vector can be found by:

$$\cos\theta_j = \frac{\underline{\mathbf{r}} \cdot \underline{\mathbf{r}}_{LOS_j}}{\|\underline{\mathbf{r}}\| \cdot \|\underline{\mathbf{r}}_{LOS_j}\|} \quad (14)$$

There is a cone about this vector whose central angle is defined by the radius of the blockage sphere and the distance from the receiver to the ISS where GPS signals are considered to be blocked. The vector from the receiver to the ISS can be found by:

$$\underline{\mathbf{r}}_{LOS_{ISS}} = \underline{\mathbf{r}}_{ISS} - \underline{\mathbf{r}} \quad (15)$$

Then the central angle, $\lambda$ can be determined using:

$$\tan(\lambda) = \frac{d/2}{\|\underline{\mathbf{r}}_{LOS_{ISS}}\|} \quad (16)$$

If the angle between the GPS line of sight vector and the ISS line of sight vector is less than $\lambda$, the signal would be blocked. Furthermore, any GPS signals below a 10° minimum elevation angle from the horizonal plane, which is perpendicular to the antenna boresight vector, are assumed to be blocked.

A side view of the GPS signal blockage model is shown in Figure 1 below. The shaded areas represent the regions where the GPS signals would be blocked.

### ISS Multipath Model

For spacecraft operating in the vicinity of the ISS, GPS signals can be blocked or degraded by multipath signals being reflected off of the ISS. It is difficult to

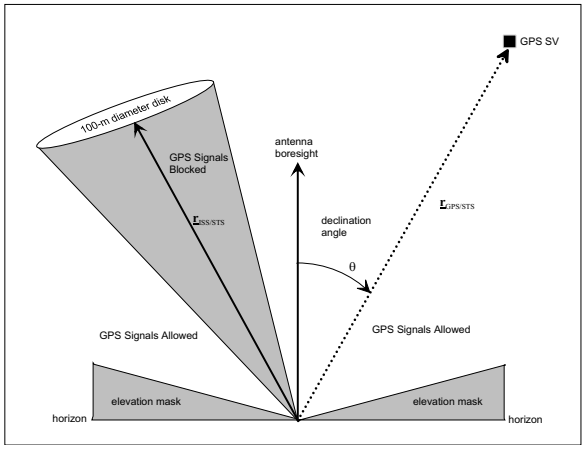


**Fig. 1 GPS Signal Blockage Model.**

model the effects of these multipath signals because the ISS is composed of several reflective surfaces, some of which are moving relative to the ISS main body. Furthermore, the chaser spacecraft is moving relative to the ISS and both are moving relative to the GPS satellites. A geometrical multipath model would have to account for each reflecting surface and the relative motion between the ISS, chaser spacecraft and GPS satellites. This would be computationally intensive and not practical for use in an integrated GPS/INS navigation simulation of a rendezvous scenario. Therefore, a statistical multipath model was selected instead of a geometrical multipath model.

The ISS multipath model computes the errors due to multipath in GPS carrier phase measurements only. The capability to compute the errors in GPS code measurements will be incorporated in the future. Therefore, the ISS multipath model is not used in the rest of this study but is included to present the current status of this work.

In formulating this model, the following assumptions are made:

1. If a GPS signal is not blocked by the ISS, it is subject to multipath.

2. For each GPS signal that is not blocked by the ISS, many reflections are caused and there is no dominant reflector.

3. The phases of the reflections are uniformly distributed over [0, $2\pi$). The rationale for this assumption is explained later in the paper.

4. The relative velocity between the chaser spacecraft and ISS is small so that there is no relative Doppler effect between the direct and reflected signals.

*GPS Carrier Phase Error*

The multipath channel consists of multiple paths or rays that have real positive gains $\beta_k$, propagation delays $\tau_k$ and phase shifts $\psi_k$, relative to the direct signal and where $k$ is the path index and in principle ranges from 0 to $\infty$ but will be limited to a finite number of rays, $N$. According to Axelrad, et al, the error in the carrier phase measurement, $\delta\phi$, due to multipath can be found by:[11]

$$\tan\delta\phi = \frac{\sum_{k=1}^{N} A_0\,\beta_k \sin\psi_k}{A_0\,\beta_0 + \sum_{k=1}^{N} A_0\,\beta_k \cos\psi_k} \tag{17}$$

Factoring out $A_0$ leaves:

$$\tan\delta\phi = \frac{\sum_{k=1}^{N} \beta_k \sin\psi_k}{\beta_0 + \sum_{k=1}^{N} \beta_k \cos\psi_k} \tag{18}$$

where $A_0$ represents the amplitude of the signal transmitted by the GPS satellite.

*Conjectures*

Since there is limited experimental data available, some conjectures based on terrestrial multipath models and measurements have been used. Conjectures about the nature of the relative phase shifts, relative amplitudes, the multipath power delay profile and relative time delays are described in this section.

*1. Phase Shifts:* The phase angle $\psi_k$ changes by $2\pi$ when the path length changes by one wavelength. For the GPS L1 signal, $f_c$ =1575.42 MHz, the wavelength is about 19 cm. This implies that small motions of the reflector or receiver can cause $\psi_k$ to change by $2\pi$. The delays associated with different paths are expected to change at different rates and in an unpredictable or random manner. If one considers a fixed transmitter and a mobile receiver, and imagines an ensemble of receiver positions spread over hundreds or thousands of wavelengths, then the geometry of a single path with delay $\tau_k$ will lead to a uniform distribution of phase for that path, while the geometrical relationship between separate paths with different delays will lead to a uniform joint distribution of pairs of phases, thus the phases would be independent. Therefore, the phase angles will be assumed to be statistically independent random variables with a uniform distribution over [0, $2\pi$).[12]

*2. Amplitudes:* The received multipath signals can be modelled as random processes. When there are a large number of paths (in practice, greater than 6), the central limit theorem may be applied so that $r\,(t)$ may be modelled as a complex-valued Gaussian random process.[12]

The received signal can be broken down into in-phase and quadrature components, $I(t)$ and $Q(t)$, which are independent Gaussian processes. This means that they are completely characterized by their mean value and autocorrelation function. $I(t)$ and $Q(t)$ have equal variance $\sigma^2$ equal to the mean square power. The total amplitude of the signal is the square root of the sum of the squares of $I(t)$ and $Q(t)$, which

are Gaussian. This leads to the conjecture that the amplitudes are Rayleigh distributed. Therefore, the $\beta_k$'s are Rayleigh distributed such that:

$$p\left(\beta_k^2\right) = \frac{1}{\bar{\beta}_k^2} e^{-\left(\beta_k^2/\bar{\beta}_k^2\right)} \tag{19}$$

where $\bar{\beta}_k^2$= the average power gain at $\tau_k$.[13]

*3. Power Delay Profile:* The multipath power-delay profile for a given environment is the expected power received as a function of delay. Numerous measurements of the multipath power-delay profile for various environments have been made, such as those by Van Rees[14] and Belloul.[15] Based on these studies, a general model of the multipath average power delay profile can be given as:[16]

$$\bar{P}\left(\tau\right) = P_0\, e^{-\tau/\bar{\tau}} \tag{20}$$

where $\bar{\tau}$ is the mean excess delay of the multipath reflections and $P_0$ is the total multipath power. The total multipath power can be estimated by using the bi-static radar equation:

$$P_0 = \frac{A_{RCS}\, \lambda^2 G_r^{mp} P_t\, G_t}{\left(4\pi\right)^3 r^2\, r_{GPS/ISS}^2} \tag{21}$$

where $A_{RCS}$ is the radar cross sectional area of the ISS, $\lambda$ is the wavelength, $G_r^{mp}$is the antenna gain of the receiver in the direction of the multipath, and $P_t\, G_t$ is the effective isotropic radiated power from the transmitter.

The average power in each multipath signal is:

$$\bar{P}_k = A_0^2\, \bar{\beta}_k^2 \tag{22}$$

Equating this with equation (20) and solving for $\bar{\beta}_k^2$ leads to:

$$\bar{\beta}_k^2 = \frac{P_0}{A_0^2}\, e^{-\tau/\bar{\tau}} \tag{23}$$

Substituting (21) into (23) and recognizing that $A_0^2 = P_t\, G_t$ results in the following expression:

$$\bar{\beta}_k^2\left(\tau\right) = \frac{A_{RCS}\, \lambda^2 G_r^{mp}}{\left(4\pi\right)^3 r^2\, r_{GPS/ISS}^2}\, e^{-\tau/\bar{\tau}} \tag{24}$$

Since the multipath rays are being reflected off of the ISS, the mean excess delay is approximated by:

$$\bar{\tau} = \frac{r}{c} \tag{25}$$

where $r$ is the distance from the spacecraft to the ISS and $c$ is the speed of light. The power of the direct signal can be estimated by using the Friis equation:

$$P_{direct} = A_0^2\, \beta_0^2 = \frac{\lambda^2 G_r^{direct} P_t\, G_t}{\left(4\pi r_{GPS/STS}\right)^2} \tag{26}$$

Since $A_0^2 = P_t\, G_t$,

$$\beta_0^2 = \frac{\lambda^2 G_r^{direct}}{\left(4\pi r_{GPS/STS}\right)^2} \tag{27}$$

If a cardioid pattern antenna is used, then:

$$G_r^{direct} = 1 + \cos\theta_{direct} \tag{28}$$

where $\theta_{direct}$ is the angle between the direct line of sight vector and the antenna boresight vector. Substituting (28) into (27) leads to:

$$\beta_0 = \sqrt{\frac{\lambda^2\left(1 + \cos\theta_{direct}\right)}{\left(4\pi r_{GPS/STS}\right)^2}} \tag{29}$$

*4. Delay Times:* The mean excess delay and RMS delay spread are commonly used to characterize multipath time delays. The parameters are determined from a multipath power delay profile. The mean excess delay is the first moment of the power delay profile and is defined to be[17]

$$\bar{\tau} = \frac{\sum\limits_k \beta_k^2\, \tau_k}{\sum\limits_k \beta_k^2} \tag{30}$$

Turin, et al, proposed that the delay times form a Poisson sequence. The probability distribution of time delays is given by:[18]

$$p\left(\tau_k\right) = \frac{1}{\bar{\tau}} e^{-\frac{\tau_k}{\bar{\tau}}} \tag{31}$$

*Multipath Model Algorithm*

This section describes how the multipath error for each GPS carrier phase measurement is computed for the simulation. The algorithm is:

1. For each simulation time and each GPS satellite, compute: $r$, $r_{GPS/ISS}$ and $\theta_{direct}$.
2. Compute $\beta_0$ using (29) and $\bar{\tau}$ using (25).
3. Given $N$, obtain the $\tau_k$'s from the Poisson distribution given in (31).
4. For each $\tau_k$:
   - Given $A_{RCS}$, compute $\bar{\beta}_k^2$ using (24).
   - Obtain the $\beta_k$'s from the Rayleigh distribution given in (19).
   - Obtain the $\psi_k$'s from a uniform distribution over [0, $2\pi$).
   - Construct $\beta_k \sin\psi_k$ and $\beta_k \cos\psi_k$
5. Determine the carrier phase error from (18).

## ISS Blockage Study Results

This section provides a quantitative analysis of this GPS signal blockage due to the ISS. A computer simulation was developed to study the GPS signal blockage due to the ISS.

The ISS orbit model was an unperturbed two-body orbit with orbit elements presented in Table 1. To determine the GPS signal blockage, the spacecraft was positioned so that it remained at a constant distance

**Table 1 ISS Orbit Elements.**

| | | |
|---|---|---|
| a | = | 6678.0 km |
| e | = | 0.005 |
| i | = | 56.0° |

**Table 2 GPS Signal Outage Statistics (All-In-View Receiver).**

| Meters Below ISS | Percent Outage | Max Outage Duration (s) | Ave. Outage Duration (s) |
|---|---|---|---|
| 10 | 99.99 | 58059 | 43197 |
| 20 | 85.85 | 2111 | 501.2 |
| 30 | 42.04 | 1119 | 167.4 |
| 40 | 12.79 | 602 | 107.3 |
| 50 | 4.92 | 389 | 103.6 |
| 60 | 2.38 | 249 | 102.8 |
| 100 | 0.0 | 0.0 | 0.0 |

$\Delta r$ directly below the ISS along the radius vector from the center of the earth to the ISS. $\Delta r$ was varied for each simulation run to determine the GPS signal blockage at different distances below the ISS. While this does not represent a rendezvous scenario, it allowed a large number of samples to be collected over the course of the simulation while maintaining the same geometry relative to the ISS. Another reason for placing the spacecraft at various distances below the ISS is to evaluate the blockage at various points during an R-bar approach in which the Space Shuttle approaches the ISS along the Earth to ISS radius vector.

Two kinds of GPS receivers were modelled. The first kind was an all-in-view receiver that was assumed to track all visible GPS satellites with no delay in tracking a satellite as soon as it becomes visible. The second kind was a 12-channel receiver programmed to track the twelve highest elevation SVs. For this receiver, it was also assumed to have no delay in tracking a satellite as soon as it becomes visible.

The simulation was run over a time span of one day taking samples once per second for the following values of $\Delta r$: 10 m, 20 m, 30 m, 40 m, 50 m, 60 m, and 100 m. At each point in time, the number of visible GPS satellites was recorded and analyzed. Any time there were less than four GPS satellites visible, it was considered to be an outage. The data collected on outages for the all-in-view receiver is summarized in Table 2.

The data shows that at least four GPS satellites were in view at all times when the satellite was 100 m or more below the ISS. At 60 m below the ISS, there were less than four satellites in view for a small percentage of the time but the average outage was over 102 seconds long. The amount of blockage increases as the satellite is brought closer to the ISS. When it is 10 m below the ISS, no position fixing is possible. These results suggest that some aiding of GPS is required when a satellite is within 60 m of the ISS.

The 12-channel receiver results were almost identical to the all-in-view receiver results. The only difference was that the number of satellites below the horizon mask was higher for the all-in-view receiver. Therefore, the performance of a 12-channel receiver programmed to select the 12 highest elevation SVs was the same as an all-in-view receiver.

## GPS/INS Integration

The integration of GPS and INS is achieved using a extended Kalman filter (EKF). This section describes the integration architecture and Kalman filter design chosen for this application.

### Integration Architecture

The complementary filter method of integration with the INS providing the reference trajectory was chosen for a number of reasons. The INS was chosen to provide the reference trajectory because it is self-contained, continuous, and provides accurate position, velocity and attitude in the short term.

The complementary filter estimates the errors in the INS position, velocity and attitude so that the linear INS error model can be used. It also allows the addition of other sensors if needed. Finally, the complementary filter can be run at a lower update rate because the dynamics of the INS errors have a much lower frequency than the dynamics of the total state variables.

Many GPS/INS integrations are feedforward or open-loop configurations where the estimates of the INS errors are not used to correct the INS. In this configuration, a linearized Kalman filter is used. The reference trajectory and actual trajectory may diverge to the point where the assumption of linearity and the random process models gradually deteriorate.[19]

To avoid this problem, a feedback or closed-loop configuration has been chosen. The estimates of the INS error are used to correct the INS, which allows the filter to be used for long durations. The closed-loop configuration requires an extended Kalman filter formulation.

### Filter Equations and Models

The general state and measurement models are given by:

$$\begin{aligned} \dot{\mathbf{x}}(t) &= \mathbf{f}\left(\mathbf{x}(t),\ t\right) + G\left(t\right)\mathbf{w}\left(t\right) \\ \mathbf{z}_k &= \mathbf{h}\left(\mathbf{x}_k\right) + \mathbf{v}_k \end{aligned} \tag{32}$$

where $\mathbf{w}\left(t\right) \sim N\left(0,\ Q\left(t\right)\right)$ and $\mathbf{v}_k \sim N\left(0,\ R_k\right)$.

In order to linearize the state and measurement models, define the output and system matrices as:

$$H_k \equiv \left.\frac{\partial \mathbf{h}}{\partial \mathbf{x}}\right|_{\mathbf{x}=\hat{\mathbf{x}}} \qquad F \equiv \left.\frac{\partial \mathbf{f}}{\partial \mathbf{x}}\right|_{\mathbf{x}=\hat{\mathbf{x}}} \tag{33}$$

The state and covariance propagation equations are:

$$\begin{aligned} &\dot{\mathbf{x}}(t) = \mathbf{f}\left(\mathbf{x}(t),\ t\right) \\ &\dot{\Phi}\left(t_k,\ t_{k-1}\right) = F\left(t_{k-1}\right)\Phi\left(t_k,\ t_{k-1}\right) \\ &P_k^- = \Phi\left(t_k,\ t_{k-1}\right)P_{k-1}^+\Phi^T\left(t_k,\ t_{k-1}\right) + Q_k \end{aligned} \tag{34}$$

**Table 3 GPS/INS Filter States**

| | |
|---|---|
| ECI Position Error | 3 |
| ECI Velocity Error | 3 |
| Attitude Quaternion Error | 4 |
| Gyro Biases | 3 |
| Accelerometer Biases | 3 |
| GPS Receiver Clock Bias | 1 |
| GPS Receiver Clock Drift | 1 |

The Kalman gain is computed by:

$$K_k = P_k^- H_k^T \left[ H_k P_k^- H_k^T + R_k \right]^{-1} \tag{35}$$

The measurement update equations are:

$$\begin{aligned} \hat{\mathbf{x}}_k^+ &= \hat{\mathbf{x}}_k^- + K_k \left[ \mathbf{y}_k - \mathbf{h}\left(\hat{\mathbf{x}}_k^-\right) \right] \\ P_k^+ &= \left[ I - K_k H_k \left(\hat{\mathbf{x}}_k^-\right) \right] P_k^- \end{aligned} \tag{36}$$

*State and Propagation Models*

The states to be estimated by the extended Kalman filter are shown in Table 3 below.

The state vector is defined as:

$$\delta\mathbf{x} = \begin{bmatrix} \delta\mathbf{r} \\ \delta\mathbf{v} \\ \delta q \\ \delta b_g \\ \delta b_a \\ \delta b_c \\ \delta d_c \end{bmatrix}_{18\times 1} \tag{37}$$

The linear system matrix is:

$$F = \begin{bmatrix} 0_{3\times3} & I_{3\times3} & 0_{3\times4} & 0_{3\times3} & 0_{3\times3} & 0_{3\times1} & 0_{3\times1} \\ \underline{G}_{3\times3} & 0_{3\times3} & D_{3\times4} & 0_{3\times3} & C_B^I(\hat{q}) & 0_{3\times1} & 0_{3\times1} \\ 0_{4\times3} & 0_{4\times3} & \Omega(\hat{\omega}) & Q(\hat{q}) & 0_{4\times3} & 0_{4\times1} & 0_{4\times1} \\ 0_{3\times3} & 0_{3\times3} & 0_{3\times3} & F_{b_g} & 0_{3\times3} & 0_{3\times1} & 0_{3\times1} \\ 0_{3\times3} & 0_{3\times3} & 0_{3\times3} & 0_{3\times3} & F_{b_a} & 0_{3\times1} & 0_{3\times1} \\ 0_{1\times3} & 0_{1\times3} & 0_{1\times4} & 0_{1\times3} & 0_{1\times3} & 0 & 1 \\ 0_{1\times3} & 0_{1\times3} & 0_{1\times4} & 0_{1\times3} & 0_{1\times3} & 0 & 0 \end{bmatrix}$$

If gyro and accelerometer biases are modelled as first order Gauss-Markov processes, then:

$$\begin{aligned} F_{b_a} &= -\frac{1}{\tau_g} I_{3\times3} \\ F_{b_a} &= -\frac{1}{\tau_g} I_{3\times3} \end{aligned}$$

where $\tau_g$ and $\tau_a$ are the gyro and accelerometer bias correlation times.

The noise input matrix is:

$$G = \begin{bmatrix} I_{3\times3} & 0_{3\times3} & 0_{3\times4} & 0_{3\times3} & 0_{3\times3} & 0_{3\times1} & 0_{3\times1} \\ 0_{3\times3} & I_{3\times3} & 0_{3\times4} & 0_{3\times3} & C_B^I(\hat{q}) & 0_{3\times1} & 0_{3\times1} \\ 0_{4\times3} & 0_{4\times3} & I_{4\times4} & Q(\hat{q}) & 0_{4\times3} & 0_{4\times1} & 0_{4\times1} \\ 0_{3\times3} & 0_{3\times3} & 0_{3\times3} & I_{3\times3} & 0_{3\times3} & 0_{3\times1} & 0_{3\times1} \\ 0_{3\times3} & 0_{3\times3} & 0_{3\times3} & 0_{3\times3} & I_{3\times3} & 0_{3\times1} & 0_{3\times1} \\ 0_{1\times3} & 0_{1\times3} & 0_{1\times4} & 0_{1\times3} & 0_{1\times3} & 0 & 1 \\ 0_{1\times3} & 0_{1\times3} & 0_{1\times4} & 0_{1\times3} & 0_{1\times3} & 0 & 0 \end{bmatrix}$$

**Table 4 INS Sensor Characteristics.**

| Parameter | | Value | Units |
|---|---|---|---|
| Gyro Bias | $\sigma$ | 0.0035 | deg/hr |
| Gyro Bias | $\tau$ | 1 | hr |
| Gyro Noise | | 0.002 | deg/$\sqrt{hr}$ |
| Accelerometer Bias | $\sigma$ | 30 | $\mu g$ |
| Accelerometer Bias | $\tau$ | 1 | hr |
| Accelerometer Noise | | 0.00075 | m/s/$\sqrt{hr}$ |

*Measurement Model*

The measured range from the $j$th GPS satellite to the vehicle at time $t_k$ can be represented by the following non-linear equation:

$$z_j(t_k) = h_j(\mathbf{x}, t_k) + v(t_k) \tag{38}$$

where $v(t_k)$ is a zero-mean, white Gaussian noise process of strength $R$ representing random range measurement noise and:

$$h_j(\mathbf{x}, t_k) = \left\| \mathbf{r}_{LOS_j}(\mathbf{x}, t_k) \right\| + \left(1 - \frac{\dot{\rho}}{c}\right) b_c \tag{39}$$

where:

$$\dot{\rho} = \frac{\mathbf{r_{LOS_j}} \cdot \dot{\mathbf{r}}_{\mathbf{LOS_j}}}{\|\mathbf{r_{LOS_j}}\|} \tag{40}$$

The linearized measurement presented to the filter is:

$$\delta z_j(t_k) = z_j(t_k) - h_j(\hat{\mathbf{x}}, t_k) \tag{41}$$

## Results

A simulation of the GPS/INS extended Kalman filter described in the preceding sections was developed. Three test cases were run. During test cases 1 and 2, the spacecraft was in an unperturbed two-body orbit with orbit elements given in Table 1 and the ISS blockage and multipath models were not used. Test case 1 was a nominal filter run where GPS measurements are continually fed into the filter. In test case 2, GPS measurements were completely blocked out for a period of 600 seconds. In test case 3, the spacecraft was started on a trajectory designed to achieve rendezvous with the ISS in one orbit period. The ISS blockage model was included in test case 3 to illustrate the effects of GPS signal blockage by the ISS on filter performance.

In all test cases, the parameters presented in Table 4 were used to simulate the gyros and accelerometers of the SIGI.[6]

In all of the test cases, the GPS/INS EKF was started with initial position errors of +10 meters in each axis and velocity errors of +0.1 m/s in each axis from the true trajectory. GPS measurements were provided at a rate of once per second for each GPS satellite above a 10-degree minimum elevation mask, except when they were blocked. The EKF dynamics model and measurement model matched the truth model. The true gyro, accelerometer and clock biases and clock drift rate were started at zero. The EKF

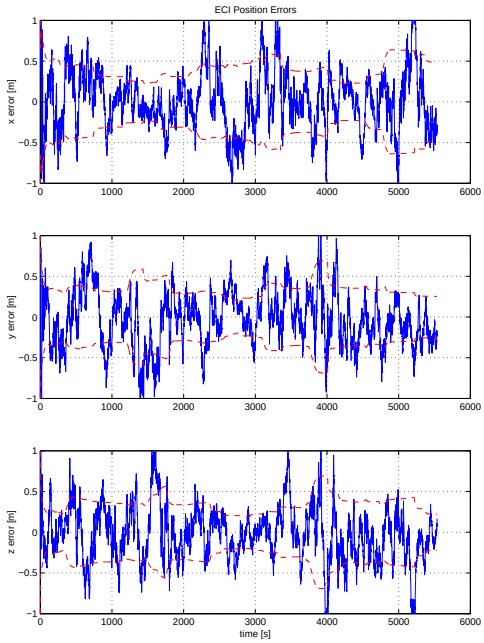


**Fig. 2 Test Case 1 ECI Position Errors.**

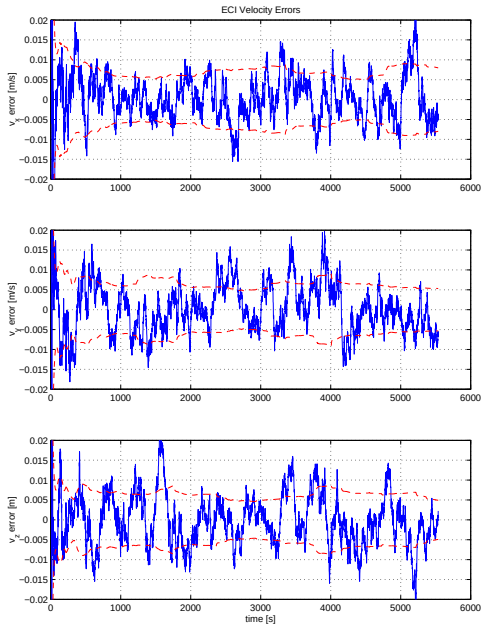


**Fig. 3 Test Case 1 ECI Velocity Errors.**

was also initialized with zero gyro, accelerometer and clock biases and clock drift rate. The simulation was run for one orbit period.

### Test Case 1 Results

The resulting position and velocity errors for test case 1 are shown in Figures 2 and 3. The red dashed lines represent the EKF estimated sigmas. The position errors were generally below ±1 meter in each axis and the velocity errors were generally below ±0.02 meters/second.

The errors in the EKF estimate of the GPS receiver clock bias were generally within ±1 meter of the true

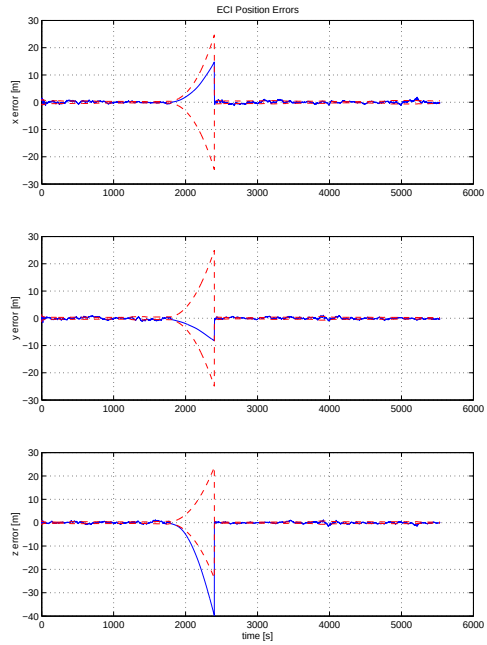


**Fig. 4 Test Case 2 ECI Position Errors.**

clock when scaled to meters and the clock drift rate errors were under ±0.5 m/s most of the time. The errors in the EKF estimate of the gyro biases for test case 1 were generally within $\pm 2.0 \times 10^{-7}$ radians/second of the true gyro biases. The errors in the EKF estimate of the accelerometer biases were generally within $\pm 2.0 \times 10^{-4}$ m/s$^2$ of the true accelerometer biases. The test case 1 results indicate that the EKF is working as expected. The sigmas track the errors which indicates that the EKF is reasonably well-tuned.

### Test Case 2 Results

During test case 2, GPS measurements were completely blocked out starting from 1800 seconds to 2400 seconds simulation time. The ECI position errors are shown in Figure 4. During the GPS outage, the total position error increases to 47.5 meters before GPS measurements reduce them to about 2 meters. The ECI velocity errors are shown in Figure 5. During the GPS outage, the total velocity error increases to 0.13 m/s just before GPS measurements are resumed. These results are very similar to those found by Um.[6]

### Test Case 3 Results

In test case 3, the velocity of the initial orbit was modified to achieve a rendezvous with the ISS within one orbit period. The spacecraft started out in the same orbit as the ISS but 15 km behind the ISS. The rendezvous trajectory is shown in Figure 6. The ISS blockage model was used during this test case.

The number of visible GPS satellites is shown in the bottom plot of Figure 9. The effects of signal blockage due to the ISS begin at about 4500 seconds simulation time when the number of visible satellites goes down to six or less. More test cases would have to be run

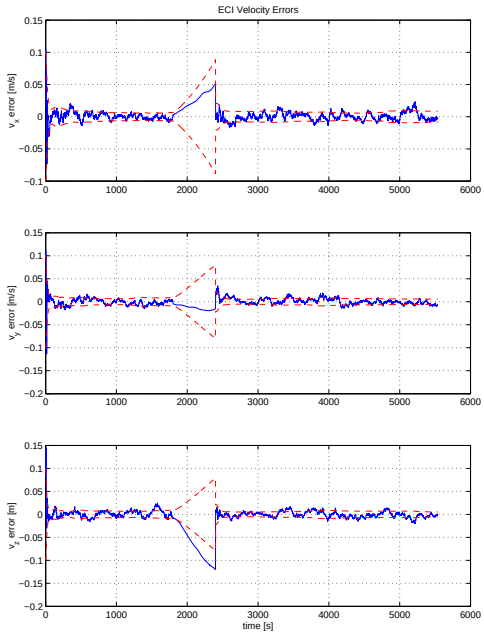


**Fig. 5 Test Case 2 ECI Velocity Errors.**

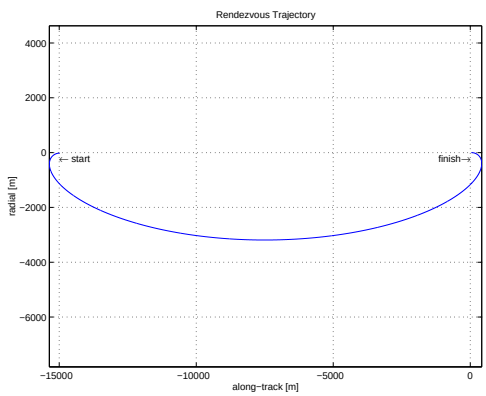


**Fig. 6 Test Case 3 Rendezvous Trajectory.**

to determine whether this is a representative case because the GPS constellation geometry may have been unusually favorable.

The ECI position errors are shown in Figure 7 and the ECI velocity errors are shown in Figure 8. The clock bias and drift estimate errors are shown in Figure 9. The filter performance does not appear to be degraded significantly by the GPS signal blockage.

## Conclusions

Error models for INS and GPS navigation sensors operating in the vicinity of the ISS have been developed. The GPS error model includes the effects of GPS signal blockage and multipath in the vicinity of the ISS. These error models have been used to form an integrated GPS/INS EKF. A simulation of the filter has been developed and the first test case shows position errors of less than 1 meter and velocity errors of less than 0.02 m/s in each axis. The second test case shows the position errors grow to approximately 47.5 meters and the velocity errors grow to 0.13 m/s

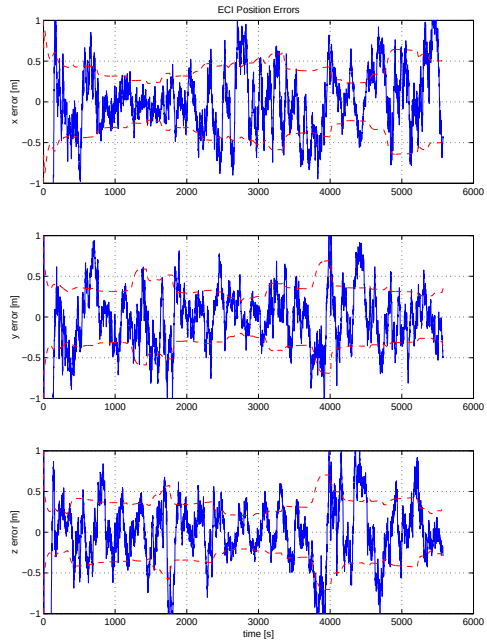


**Fig. 7 Test Case 3 ECI Position Errors.**

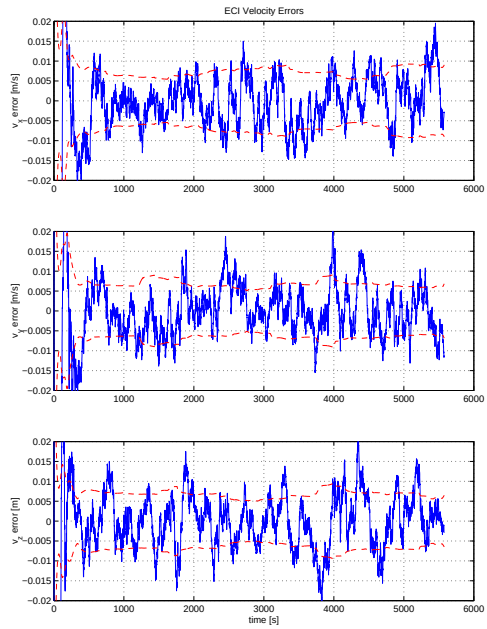


**Fig. 8 Test Case 3 ECI Velocity Errors.**

during a ten minute GPS outage. Finally, the third test case shows that the GPS/INS EKF performance is not significantly degraded by GPS signal blockage due to the ISS during a simulated rendezvous with the ISS.

### Future Work

Plans for the future include adding the effects of GPS satellite clock and ephemeris errors and more realistic orbital dynamics models to the simulation. The GPS/INS EKF will be modified to perform relative navigation with the ISS using GPS carrier phase measurements. The ISS multipath model will then be

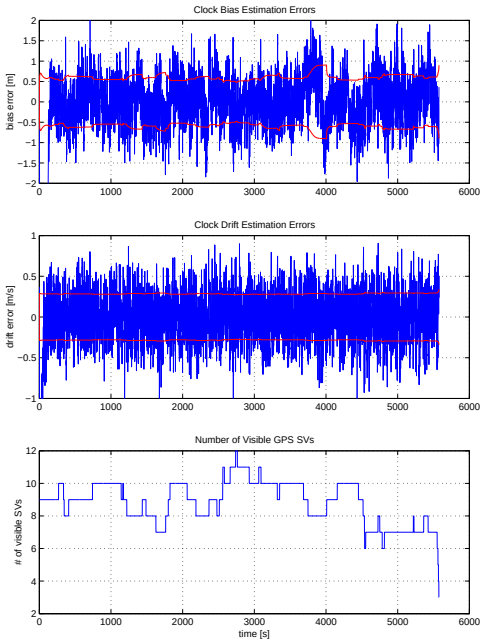


**Fig. 9 Test Case 3 Clock Bias, Clock Drift Estimation Errors and Number of Visible GPS Satellites.**

incorporated and the effects of signal blockage combined with multipath will be studied.

## Acknowledgments

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